CV

Curriculum vitae. This page mirrors the substantive content of my academic CV in searchable, selectable HTML.

artsmorris@gmail.com · Google Scholar · arXiv

Education

Oct 2021–Sep 2025 **PhD in Theoretical Condensed Matter Physics**
King's College, University of Cambridge · Cambridge, England

- EPSRC-funded PhD in the Theory of Condensed Matter group, supervised by Robert-Jan Slager.
- Research on topological phases of matter, especially multiband Euler materials.

Oct 2020–Jun 2021 **Master's in Mathematical and Theoretical Physics (MMathPhys)**
Corpus Christi College, University of Oxford · Oxford, England

- Distinction — 77.7%.

Oct 2017–Oct 2020 **BA in Physics**
Corpus Christi College, University of Oxford · Oxford, England

- Part B examinations: First Class — 76.1%, 10th/125 in cohort.
- Part A examinations: First Class — 86.3%, 11th/159 in cohort.
- Preliminary examinations: Distinction — 82.6%, 18th/178 in cohort.

Publications

1. Constraints on phantom codes from automorphism group bounds — A. S. Morris, D. Malz. arXiv:2604.15111 (2026); accepted in *Physical Review Letters*.
2. Euler topology in classical spin liquids — L. Rüegg, A. S. Morris, H. Yan, R.-J. Slager. arXiv:2505.09683 (2025).
3. Optical manifestations and bounds of topological Euler class — W. J. Jankowski, A. S. Morris, A. Bouhon, F. N. Ünal, R.-J. Slager. *Physical Review B* 111, L081103 (2025).
4. Non-Abelian Hopf-Euler insulators — W. J. Jankowski*, A. S. Morris*, Z. Davoyan, A. Bouhon, F. N. Ünal, R.-J. Slager. *Physical Review B* 110, 075135 (2024). *Equal contribution.
5. Andreev reflection in Euler materials — A. S. Morris, A. Bouhon, R.-J. Slager. *New Journal of Physics* 26, 023014 (2024).
6. Phasik: a Python package to identify system states in partially temporal networks — M. Lucas, A. Townsend-Teague, M. Neri, S. Poetto, A. S. Morris, B. Habermann, L. Tichit. *Journal of Open Source Software* 8(91), 5872 (2023).
7. Inferring cell cycle phases from a partially temporal network of protein interactions — M. Lucas, A. S. Morris, A. Townsend-Teague, L. Tichit, B. Habermann, A. Barrat. *Cell Reports Methods* 3, 100397 (2023).

Experience

Oct 2025–present **Postdoctoral researcher · Malz group, QMATH**
University of Copenhagen · Copenhagen, Denmark

- Funded by a DeiC grant.
- Working on quantum error correction.

17 Jun–27 Aug 2024 **JSPS Summer Scholar · Oka group**
Institute for Solid State Physics, University of Tokyo · Kashiwa, Japan

- Collaborated on projects related to classical spin liquids and the topology of chemical reaction networks.

18 Aug–17 Dec 2023	**Winton-Kavli Cambridge-Berkeley Exchange Scholar · Moore group** UC Berkeley · Berkeley, California, USA • Researched the use of Green's functions for the bulk-boundary correspondence in multiband topological quantum phases.
Dec 2021–present	**Associate · NanoDTC** Cambridge, UK • Participate in journal clubs and events spanning energy materials, nanoelectronics, nanophotonics, sustainable nano and bio-nanotechnologies.
30 Jun–30 Jul 2020	**Summer researcher · Computational Biology (Habermann) group, CENTURI** Marseille, France · completed remotely • Used Python for gene-expression data analysis and clustering to study time-dependent protein-interaction networks.
1 Jul–10 Aug 2019	**Oxford University Internship Award · Fluid dynamics summer school** Perm State University · Perm, Russia • Courses in fluid dynamics, programming for fluid dynamics and parallel programming with MPI. • Used Fortran 90 and 77 to write fluid simulations from scratch.
1–21 Sep 2018	**Kathleen Lavidge Bursary** Stanford University · Palo Alto, California, USA • Course on "The Intersection of Art and Science"; achieved an A+ in all areas.
19 Jun–18 Jul 2018	**Expanding Horizons Scholar · CLASSE REU, Wilson Synchrotron Laboratory** Cornell University · Ithaca, New York, USA • Reviewed methods for optimising collimation in accelerator beamlines, published internally as CBETA technical note 27.
24 Jun–8 Jul 2016	**Summer-school participant** CERN · Geneva, Switzerland • Joined undergraduate summer-school students in lectures on accelerator physics, the Standard Model and related topics.

Awards and prizes

2024	JSPS Summer Scholar
2023	Winton-Kavli Graduate Exchange Scholar
2021–2024	NanoDTC Associate
2021–2024	EPSRC PhD Studentship
2018, 2019, 2020, 2021	Corpus Christi Academic Scholarship for performance in university examinations
2021	Palmer Scholarship for European travel
2020	Oxford Physics Department prize for practical work
2018, 2019, 2021	Corpus Christi travel grants
2019	Oxford University Internship Award
2018	Kathleen Lavidge Bursary
2018	Expanding Horizons Scholarship

Presentations

29 Jan 2025	QLunch, QMATH, University of Copenhagen — 1-hour talk
19 Aug 2024	Seminar, Quantum Matter Group, Okinawa Institute of Science and Technology — 1-hour talk
7 Aug 2024	Seminar, Yukawa Institute, University of Kyoto — 1-hour talk
3 Jun 2024	Seminar, Institute of Solid State Physics, University of Tokyo — 1-hour talk
19 Jun 2024	JSPS Orientation, SOKENDAI — poster
29 May 2024	Collective Phenomena Group Meeting, University of Cambridge — 1-hour talk
30 Nov 2023	TI meeting, University of Pennsylvania — 1-hour talk
6 Sep 2023	Quantum Seminar, UC Berkeley — 1-hour talk

Conferences and summer schools

2–4 Jun 2026	QMATH/Qusoft

joint outing · Hillerød, Denmark

20–24 May 2026	DeiC quantum algorithms workshop

· Sønderborg, Denmark

24–30 Apr 2026	Monte Verità Quantum Codes

· Ascona, Switzerland

24–30 Jan 2026	QIP

· Riga, Latvia

17–20 Jun 2024	JSPS Orientation

· SOKENDAI, Japan

3–7 Jun 2024	Condensed Matter in the City

· UCL, England

10–14 Jul 2023	Condensed Matter in the City

· UCL, England

22 Aug–10 Sep 2022	Quantum Dynamics: From Electrons to Qbits

· ICTP, Italy

Teaching

Jan–Jun 2026	Supervised three final-year bachelor student projects at the University of Copenhagen.
Feb–Mar 2024 & 2025	Supervised 11 (2024) and 13 (2025) Cambridge master’s students in the Phase Transitions course.

Oct 2021–Jun 2023	Supervised four groups of second-year Cambridge undergraduates (10 students total) in Mathematical Methods for Natural Sciences.
Oct 2022–present	Private tutor in A-Level mathematics and physics.

Outreach

2023	Runner-up in the East-Anglia heats of the
Institute of Physics Three Minute Wonder competition	.
Oxford	Physics subject ambassador for Corpus Christi College, helping with open days and interviews and speaking with prospective applicants and the public.